```
Analyzing seed 172180395821707403
Analyzing seed 283404770198602602
Analyzing seed 305986450614148690
Analyzing seed 1230287949579537403
Analyzing seed 1888415619847841128
Analyzing seed 1918007468868772336
Analyzing seed 2461295749352584108
Analyzing seed 3258463328937640595
Analyzing seed 3582616062465030267
Analyzing seed 3704716633526084363
Analyzing seed 3952397966055989215
Analyzing seed 4040285944958847150
Analyzing seed 4219039243791140697
Analyzing seed 4619122171158768524
Analyzing seed 4932511216876650745
Analyzing seed 5213047409853278064
Analyzing seed 5415544643585892346
Analyzing seed 6023008189364820510
Analyzing seed 6108867655028980287
Analyzing seed 6942591650975756121
Analyzing seed 9036321132460399255
Analyzing seed 9082055256280075172
Analyzing seed 9189988769602851417
Analyzing seed 10431384713483787270
Analyzing seed 10856823803646276806
Analyzing seed 11242426753702356865
Analyzing seed 12123519807951751650
Analyzing seed 12851640484293199234
Analyzing seed 14040336450255043598
Analyzing seed 14924267685654216746
Analyzing seed 15403081841477312463
Analyzing seed 15853699719446171583
Analyzing seed 16329425431442249417
Analyzing seed 16627818730624180313
Analyzing seed 16952466944611321358
Analyzing seed 17279552413280415429
Analyzing seed 17521310920850592782
Analyzing seed 17715532056777442089
Analyzing seed 17883847597205629106
Analyzing seed 18127452951968972828

AIEF mature stationarity and feasible-perturbation analysis
============================================================

Mature adaptive stationarity
----------------------------
  seed=172180395821707403: stationary_fraction=1, mean
||P_w(F)||/||F||=9.96029e-14, max=1.42818e-12, zero-set switches=0,
saturated-set switches=0
```

, max B drift=4.43115e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.7e-17, replay_exact=True
  seed=283404770198602602: stationary_fraction=1, mean
||P_w(F)||/||F||=1.75457e-13, max=2.75971e-12, zero-set switches=0,
saturated-set switches=0
, max B drift=1.72456e-11, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=4.91e-17, replay_exact=True
  seed=305986450614148690: stationary_fraction=1, mean
||P_w(F)||/||F||=1.91646e-13, max=3.23833e-12, zero-set switches=0,
saturated-set switches=0
, max B drift=4.72533e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.54e-17, replay_exact=True
  seed=1230287949579537403: stationary_fraction=1, mean
||P_w(F)||/||F||=1.18532e-12, max=2.10587e-11, zero-set switches=0,
saturated-set switches=
0, max B drift=7.27283e-11, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.67e-17, replay_exact=True
  seed=1888415619847841128: stationary_fraction=1, mean
||P_w(F)||/||F||=1.11711e-13, max=1.67395e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=2.57059e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.6e-17, replay_exact=True
  seed=1918007468868772336: stationary_fraction=1, mean
||P_w(F)||/||F||=7.78676e-14, max=9.78005e-13, zero-set switches=0,
saturated-set switches=
0, max B drift=8.10498e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.6e-17, replay_exact=True
  seed=2461295749352584108: stationary_fraction=1, mean
||P_w(F)||/||F||=4.49434e-14, max=3.30867e-13, zero-set switches=0,
saturated-set switches=
0, max B drift=9.19448e-13, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=9.81e-18, replay_exact=True
  seed=3258463328937640595: stationary_fraction=1, mean
||P_w(F)||/||F||=1.01956e-13, max=1.43812e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=3.32296e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=4.39e-17, replay_exact=True
  seed=3582616062465030267: stationary_fraction=1, mean
||P_w(F)||/||F||=8.6123e-14, max=1.05571e-12, zero-set switches=0, saturated-
set switches=0
, max B drift=3.98931e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=9.81e-18, replay_exact=True
  seed=3704716633526084363: stationary_fraction=1, mean
||P_w(F)||/||F||=1.1187e-13, max=1.57609e-12, zero-set switches=0, saturated-
set switches=0
, max B drift=2.9748e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.1e-17, replay_exact=True
  seed=3952397966055989215: stationary_fraction=1, mean
||P_w(F)||/||F||=1.08241e-13, max=1.52927e-12, zero-set switches=0,

saturated-set switches=
0, max B drift=3.79974e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.25e-17, replay_exact=True
  seed=4040285944958847150: stationary_fraction=1, mean
||P_w(F)||/||F||=7.97556e-14, max=1.1087e-12, zero-set switches=0, saturated-
set switches=0
, max B drift=3.22999e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.4e-17, replay_exact=True
  seed=4219039243791140697: stationary_fraction=1, mean
||P_w(F)||/||F||=1.24233e-13, max=2.02643e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=7.64049e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=0, replay_exact=True
  seed=4619122171158768524: stationary_fraction=0.115125, mean
||P_w(F)||/||F||=0.0569358, max=0.160206, zero-set switches=0, saturated-set
switche
s=1, max B drift=0.208939, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.54e-17, replay_exact=True
  seed=4932511216876650745: stationary_fraction=1, mean
||P_w(F)||/||F||=5.82875e-14, max=6.88792e-13, zero-set switches=0,
saturated-set switches=
0, max B drift=7.60477e-13, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.4e-17, replay_exact=True
  seed=5213047409853278064: stationary_fraction=1, mean
||P_w(F)||/||F||=6.64507e-14, max=8.08965e-13, zero-set switches=0,
saturated-set switches=
0, max B drift=1.98264e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.78e-17, replay_exact=True
  seed=5415544643585892346: stationary_fraction=1, mean
||P_w(F)||/||F||=9.39025e-14, max=1.16187e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=2.64868e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.7e-17, replay_exact=True
  seed=6023008189364820510: stationary_fraction=1, mean
||P_w(F)||/||F||=7.57295e-14, max=1.01207e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=5.55414e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.7e-17, replay_exact=True
  seed=6108867655028980287: stationary_fraction=1, mean
||P_w(F)||/||F||=9.63434e-14, max=1.19049e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=3.39442e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.1e-17, replay_exact=True
  seed=6942591650975756121: stationary_fraction=1, mean
||P_w(F)||/||F||=7.61809e-14, max=8.56888e-13, zero-set switches=0,
saturated-set switches=
0, max B drift=1.91521e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.6e-17, replay_exact=True
  seed=9036321132460399255: stationary_fraction=1, mean

```
||P_w(F)||/||F||=1.0392e-13, max=1.52455e-12, zero-set switches=0, saturated-
set switches=0
, max B drift=3.00391e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.94e-17, replay_exact=True
  seed=9082055256280075172: stationary_fraction=1, mean
||P_w(F)||/||F||=1.55613e-13, max=2.40277e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=6.67596e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=4.39e-17, replay_exact=True
  seed=9189988769602851417: stationary_fraction=1, mean
||P_w(F)||/||F||=9.71492e-14, max=1.30153e-12, zero-set switches=0,
saturated-set switches=
0, max B drift=6.86555e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.39e-17, replay_exact=True
  seed=10431384713483787270: stationary_fraction=1, mean
||P_w(F)||/||F||=7.00657e-14, max=8.55589e-13, zero-set switches=0,
saturated-set switches
=0, max B drift=2.32124e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.96e-17, replay_exact=True
  seed=10856823803646276806: stationary_fraction=1, mean
||P_w(F)||/||F||=1.25824e-13, max=1.91392e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=4.33768e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.25e-17, replay_exact=True
  seed=11242426753702356865: stationary_fraction=1, mean
||P_w(F)||/||F||=7.83726e-14, max=9.972e-13, zero-set switches=0, saturated-
set switches=0
, max B drift=2.27496e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.7e-17, replay_exact=True
  seed=12123519807951751650: stationary_fraction=1, mean
||P_w(F)||/||F||=7.8453e-14, max=1.09766e-12, zero-set switches=0, saturated-
set switches=
0, max B drift=4.25432e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.19e-17, replay_exact=True
  seed=12851640484293199234: stationary_fraction=1, mean
||P_w(F)||/||F||=4.32554e-14, max=5.32567e-13, zero-set switches=0,
saturated-set switches
=0, max B drift=1.84928e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.54e-17, replay_exact=True
  seed=14040336450255043598: stationary_fraction=1, mean
||P_w(F)||/||F||=8.80282e-14, max=1.26214e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=3.53566e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=4.16e-17, replay_exact=True
  seed=14924267685654216746: stationary_fraction=1, mean
||P_w(F)||/||F||=6.66157e-14, max=8.46693e-13, zero-set switches=0,
saturated-set switches
=0, max B drift=2.71523e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.1e-17, replay_exact=True
```

```
  seed=15403081841477312463: stationary_fraction=1, mean
||P_w(F)||/||F||=1.18361e-13, max=1.80085e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=4.44102e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.39e-17, replay_exact=True
  seed=15853699719446171583: stationary_fraction=1, mean
||P_w(F)||/||F||=1.2699e-13, max=1.99444e-12, zero-set switches=0, saturated-
set switches=
0, max B drift=7.50579e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=2.4e-17, replay_exact=True
  seed=16329425431442249417: stationary_fraction=1, mean
||P_w(F)||/||F||=7.13448e-14, max=9.72213e-13, zero-set switches=0,
saturated-set switches
=0, max B drift=3.38388e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.96e-17, replay_exact=True
  seed=16627818730624180313: stationary_fraction=1, mean
||P_w(F)||/||F||=7.39512e-14, max=1.02939e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=1.54925e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.39e-17, replay_exact=True
  seed=16952466944611321358: stationary_fraction=1, mean
||P_w(F)||/||F||=1.02441e-13, max=1.58137e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=3.57766e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.25e-17, replay_exact=True
  seed=17279552413280415429: stationary_fraction=1, mean
||P_w(F)||/||F||=7.51491e-14, max=9.64569e-13, zero-set switches=0,
saturated-set switches
=0, max B drift=2.89497e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=4.16e-17, replay_exact=True
  seed=17521310920850592782: stationary_fraction=1, mean
||P_w(F)||/||F||=1.13311e-13, max=1.65346e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=5.22557e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=1.96e-17, replay_exact=True
  seed=17715532056777442089: stationary_fraction=1, mean
||P_w(F)||/||F||=9.21112e-14, max=1.38956e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=2.59041e-12, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=3.93e-17, replay_exact=True
  seed=17883847597205629106: stationary_fraction=1, mean
||P_w(F)||/||F||=4.44722e-13, max=7.85276e-12, zero-set switches=0,
saturated-set switches
=0, max B drift=5.75374e-11, tangent/idempotence/homogeneity failures=0/0/0,
equivariance error=4.39e-17, replay_exact=True
  seed=18127452951968972828: stationary_fraction=1, mean
||P_w(F)||/||F||=1.80724e-13, max=3.0378e-12, zero-set switches=0, saturated-
set switches=
0, max B drift=4.0923e-12, tangent/idempotence/homogeneity failures=0/0/0,
```

```
equivariance error=3.93e-17, replay_exact=True

Frozen-fast restoring diagnostics
---------------------------------
  interior_mixing, delta=0.0001
    restoring_coefficient: median=1705.81, min=1705.59, max=3812.36
    restoring_cosine: median=0.670362, min=0.67027, max=0.931897
    transverse_to_field_ratio: median=0.742034, min=0.362723, max=0.742117
  interior_mixing, delta=0.001
    restoring_coefficient: median=170.594, min=170.573, max=381.249
    restoring_cosine: median=0.670392, min=0.6703, max=0.931901
    transverse_to_field_ratio: median=0.742008, min=0.362712, max=0.742091
  interior_mixing, delta=0.01
    restoring_coefficient: median=17.0729, min=17.0708, max=38.1384
    restoring_cosine: median=0.670683, min=0.670591, max=0.931945
    transverse_to_field_ratio: median=0.741744, min=0.3626, max=0.741827
  interior_mixing, delta=0.05
    restoring_coefficient: median=3.42658, min=3.42616, max=7.63969
    restoring_cosine: median=0.671977, min=0.671885, max=0.932137
    transverse_to_field_ratio: median=0.740572, min=0.362106, max=0.740655
  radial_inward, delta=0.0001
    restoring_coefficient: median=1328.18, min=1328.18, max=2685.68
    restoring_cosine: median=0.793147, min=0.793147, max=0.993161
    transverse_to_field_ratio: median=0.60903, min=0.116754, max=0.60903
  radial_inward, delta=0.001
    restoring_coefficient: median=132.831, min=132.831, max=268.581
    restoring_cosine: median=0.793177, min=0.793177, max=0.993161
    transverse_to_field_ratio: median=0.608991, min=0.116749, max=0.608991
  radial_inward, delta=0.01
    restoring_coefficient: median=13.2966, min=13.2966, max=26.8716
    restoring_cosine: median=0.793476, min=0.793476, max=0.993168
    transverse_to_field_ratio: median=0.608602, min=0.116691, max=0.608602
  radial_inward, delta=0.05
    restoring_coefficient: median=2.67133, min=2.67133, max=5.38632
    restoring_cosine: median=0.794797, min=0.794797, max=0.993198
    transverse_to_field_ratio: median=0.606876, min=0.116434, max=0.606876
  tangent_face, delta=0.0001
    restoring_coefficient: median=0.315789, min=0.221397, max=53.4947
    restoring_cosine: median=1, min=0.246348, max=1
    transverse_to_field_ratio: median=1.14333e-08, min=3.91233e-09,
max=0.969181
  tangent_face, delta=0.001
    restoring_coefficient: median=0.315789, min=-29145.1, max=0.315789
    restoring_cosine: median=1, min=-0.424179, max=1
    transverse_to_field_ratio: median=1.19457e-09, min=3.80634e-10,
max=0.905579
  tangent_face, delta=0.01
    restoring_coefficient: median=0.315789, min=-5604.16, max=0.315789
    restoring_cosine: median=1, min=-0.345925, max=1
```

```
    transverse_to_field_ratio: median=1.23926e-10, min=3.8217e-11,
max=0.938262
  tangent_face, delta=0.05
    restoring_coefficient: median=0.315789, min=0.200872, max=0.37363
    restoring_cosine: median=1, min=0.764327, max=1
    transverse_to_field_ratio: median=2.45775e-11, min=7.77796e-12,
max=0.644829

Coupled restoration diagnostics
-------------------------------
  interior_mixing, delta=0.0001
    final_to_initial_distance_ratio: median=4.27237e-12, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=0.5, valid=40/40
  interior_mixing, delta=0.001
    final_to_initial_distance_ratio: median=5.3518e-13, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=0.5, valid=40/40
  interior_mixing, delta=0.01
    final_to_initial_distance_ratio: median=3.49086e-11, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=0.5, valid=40/40
  interior_mixing, delta=0.05
    final_to_initial_distance_ratio: median=4.40604e-09, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=1, valid=40/40
  radial_inward, delta=0.0001
    final_to_initial_distance_ratio: median=5.07018e-12, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=0.5, valid=40/40
  radial_inward, delta=0.001
    final_to_initial_distance_ratio: median=4.97561e-13, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=0.5, valid=40/40
  radial_inward, delta=0.01
    final_to_initial_distance_ratio: median=4.87147e-11, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=0.5, valid=40/40
  radial_inward, delta=0.05
    final_to_initial_distance_ratio: median=6.23636e-09, valid=40/40
    time_to_50pct_distance: median=0.5, valid=40/40
    time_to_10pct_distance: median=1, valid=40/40
  tangent_face, delta=0.0001
    final_to_initial_distance_ratio: median=1.42123e-07, valid=40/40
    time_to_50pct_distance: median=2.5, valid=40/40
    time_to_10pct_distance: median=7.5, valid=40/40
  tangent_face, delta=0.001
    final_to_initial_distance_ratio: median=1.42425e-07, valid=40/40
    time_to_50pct_distance: median=2.5, valid=40/40
```

```
    time_to_10pct_distance: median=7.5, valid=40/40
  tangent_face, delta=0.01
    final_to_initial_distance_ratio: median=1.50665e-07, valid=40/40
    time_to_50pct_distance: median=2.5, valid=40/40
    time_to_10pct_distance: median=7.5, valid=40/40
  tangent_face, delta=0.05
    final_to_initial_distance_ratio: median=1.83456e-07, valid=40/40
    time_to_50pct_distance: median=2.5, valid=40/40
    time_to_10pct_distance: median=7.5, valid=40/40

Interpretation constraints
--------------------------
* Stationarity refers only to the adaptive B-component, not the full delayed
state.
* Frozen-fast restoration holds theta/history/EMA fixed for one instantaneous
field evaluation.
* Coupled restoration evolves all state variables and answers a different
stability question.
* Positive radial restoration does not exclude large transverse or non-normal
motion.
* The ideal directional field excludes the fixed denominator safeguard.
```